

Microeconomic Theory: TA Session

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Reminders

Solutions to the 2nd-half problem sets will be posted in our section's Github folder

- ▶ Refer to the “Welcome to Section 02” Canvas announcement if you don't know the link to the Github folder

Game Theory: Nash Equilibrium

Nash equilibrium: an equilibrium in which players each choose their best strategy given the strategies that all other players have chosen

- ▶ i.e. nobody has an incentive to (unilaterally) deviate from their equilibrium choice

Pure-strategy Nash equilibrium: prisoners' dilemma

The police arrested two criminals, A and B, who are suspected of committing a major crime, and **questions them separately**

- ▶ If one of them confesses and the other remain silent, the confessed one goes free, whereas the silent one gets 10 years
- ▶ If both of them confess, they both get 5 years
- ▶ If none of them confess, they both get 2 years

		B	
		Confess	Silent
A	Confess	-5, -5	0, -10
	Silent	-10, 0	-2, -2

Pure-strategy Nash equilibrium: prisoners' dilemma

		B	
		Confess	Silent
A	Confess	-5, -5	0, -10
	Silent	-10, 0	-2, -2

Nash equilibrium: (Confess, Confess)

- This is not optimal: welfare is maximized at (Silent, Silent)

Oligopoly - Cournot competition

Cournot competition: firms simultaneously decide their quantities; price is determined by the total output of the firms

Suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. The market demand curve is $P = 10 - Q$. The firms engage in Cournot competition.

1. Calculate the equilibrium outputs, price and profits.

Answer:

Let q_1, q_2 be the quantities of the two firms, so $Q = q_1 + q_2$.

The demand curve that firm 1 faces is $P = (10 - q_2) - q_1$. Then, firm 1's revenue $= Pq_1 = (10 - q_2)q_1 - q_1^2$, $MR_1 = (10 - q_2) - 2q_1$.

Profit maximization means $MR_1 = MC \Rightarrow (10 - q_2) - 2q_1 = 2$.

The same with firm 2: $MR_2 = MC \Rightarrow (10 - q_1) - 2q_2 = 2$.

Solving the above two equations, we get $q_1 = q_2 = \frac{8}{3}$,
 $P = 10 - q_1 - q_2 = \frac{14}{3}$. The profit of each firm is $(P - 2)q_1 = \frac{64}{9}$.

Oligopoly - Cournot competition

Suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. The market demand curve is $P = 10 - Q$. The firms engage in Cournot competition.

2. Suppose there are N firms in this market. Calculate the equilibrium output of each firm q_i , total output of all firms Q , the market price P , each firm's profit π_i , and total profit of all firms Π .

Answer: Denote q_{-i} as the total output of all firms except firm i . Then, the demand curve that firm i faces is $P = (10 - q_{-i}) - q_i$. Then, $MR = (10 - q_{-i}) - 2q_i$.

Profit maximization means $MR_i = MC$: $(10 - q_{-i}) - 2q_i = 2$
 $\Rightarrow q_i = \frac{8 - q_{-i}}{2}$. This is true for all i from 1 to N , so all firms have the same q_i , and $q_{-i} = (N - 1)q_i$. So, $q_i = \frac{8 - (N-1)q_i}{2} \Rightarrow q_i = \frac{8}{N+1}$.

Total output $Q = \frac{8N}{N+1}$, price $P = 10 - \frac{8N}{N+1}$, each firm's profit $\pi_i = (10 - \frac{8N}{N+1} - 2)\frac{8}{N+1} = (\frac{8}{N+1})^2$. Total profit $\Pi = N(\frac{8}{N+1})^2$.

3. As N increases to infinity, how does Q, p, π_i, Π change?

Answer: As N increases to infinity, $Q = 8, P = 2, \pi_i = 0, \Pi = 0$. The equilibrium approaches the perfect competition outcome as N increases.

Stackelberg competition

Stackelberg competition: firms sequentially decide their quantities; price is determined by the total output of the firms

Suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. The market demand curve is $P = 10 - Q$. Firm A chooses its quantity first. Then, after observing Firm 1's quantity, Firm 2 chooses its own quantity.

To solve this kind of sequential problem, we use **backward induction**

- ▶ First, we solve the profit maximization problem of Firm 2, taking q_1 as given. We get the optimal q_2 in terms of q_1 .
- ▶ Then, using this q_2 , solve the profit maximization problem of Firm 1

Stackelberg competition

Suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. The market demand curve is $P = 10 - Q$. Firm A chooses its quantity first. Then, after observing Firm 1's quantity, Firm 2 chooses its own quantity.

1. Solve for the two firms' quantity, price, and profit.

Answer: Using backward induction, solve for q_2 treating q_1 as given.

$P = 10 - q_1 - q_2$, $MR_2 = 10 - q_1 - 2q_2$. Profit maximization means $MR_2 = MC$: $10 - q_1 - 2q_2 = 2 \Rightarrow q_2^* = \frac{8 - q_1}{2}$.

Now, using this expression of q_2^* , we solve the profit maximization problem of Firm 1:

$P = 10 - q_1 - q_2 = 10 - q_1 - \frac{8 - q_1}{2} = 6 - \frac{q_1}{2}$, $MR_1 = 6 - q_1$.

$MR_1 = MC$: $6 - q_1 = 2 \Rightarrow q_1^* = 4$.

Then, we get $q_2^* = \frac{8 - q_1^*}{2} = 2$, $P = 10 - q_1^* - q_2^* = 4$.

Profit of each firm: $\pi_1 = (P - 2)q_1^* = 8$, $\pi_2 = (P - 2)q_2^* = 4$.